



BANKING CUSTOMER 360

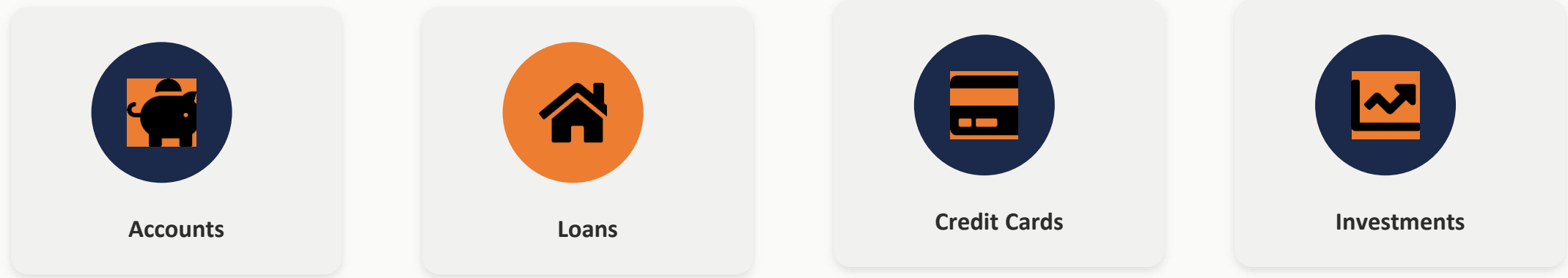
A unified, real-time view of every customer across Loans, Accounts, Credit Cards & Investments

Client : A Leading Global Bank

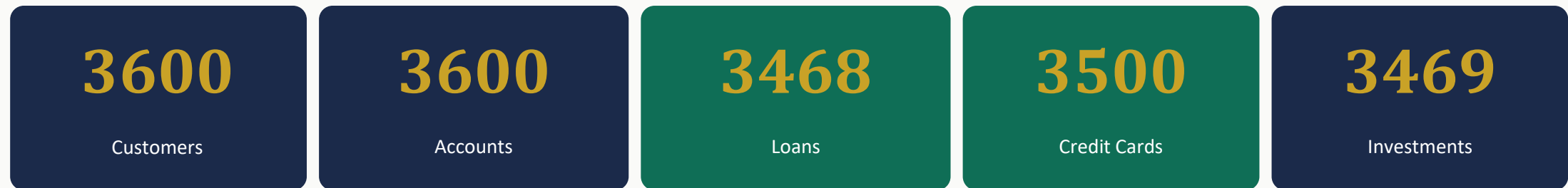
The Problem: Customer data, Fragmented

Four products. Four silos. No single view of the relationship.

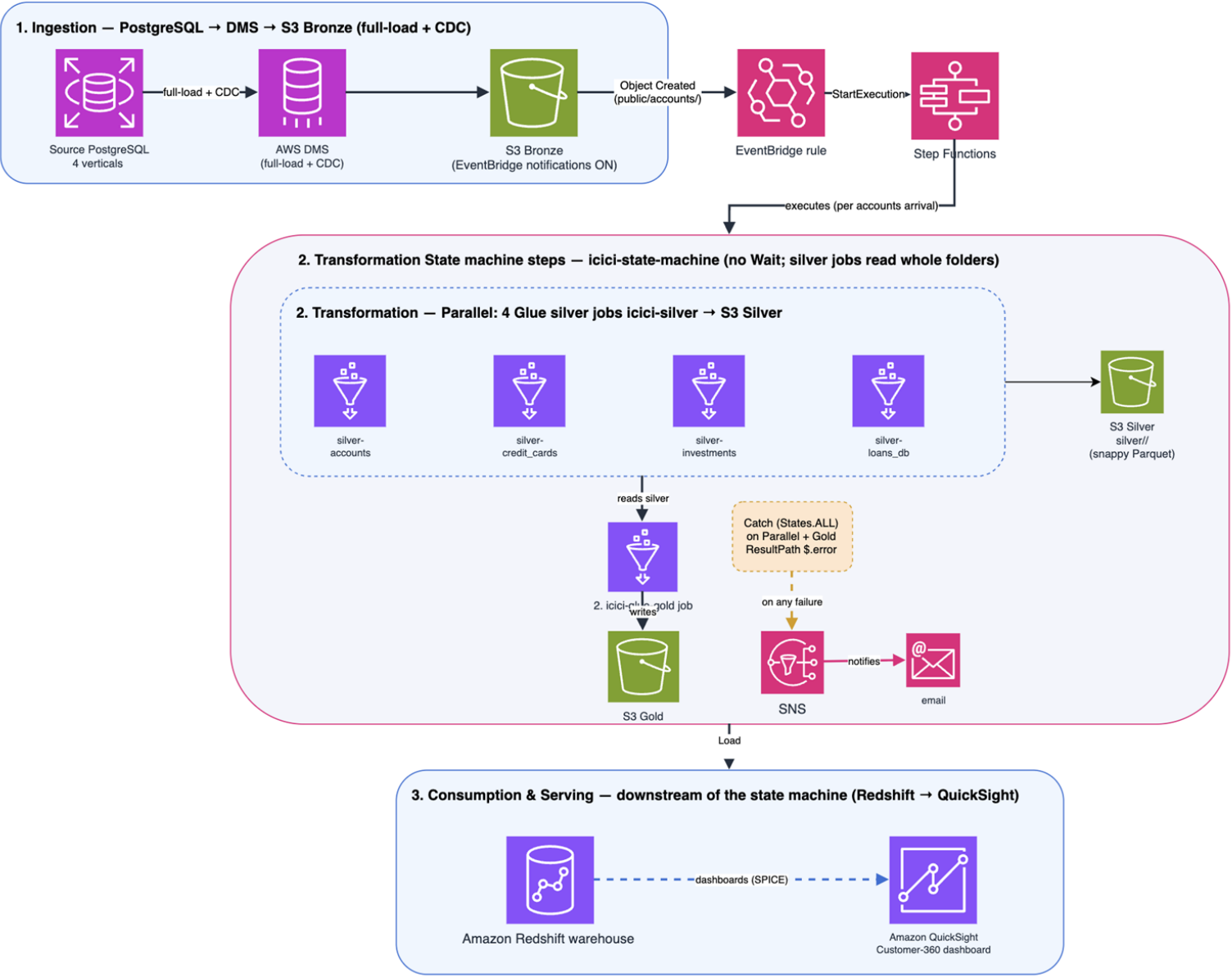
Bank can't see cross-sell opportunity, true relationship value, or regional risk



A total of **14,600** rows of synthetic (Full Load + CDC) test data was generated across the four verticals — Loans, Accounts, Credit Cards, and Investments — to validate the pipeline at scale."



PostgreSQL → DMS → S3 Bronze → EventBridge → Step Functions → 4 Silver → Gold → Redshift → QuickSight



Refining the Data: Bronze → Silver → Gold

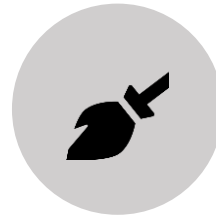
Every stage makes the data more trustworthy and more useful.



Bronze

Raw, as-is

Straight from the source —
unfiltered, unformatted,
untouched.



Silver

Cleaned & flagged

Types fixed, duplicates
removed, issues flagged —
never hidden.



Gold

Business-ready

One customer view, joined
and modeled for instant
analysis.

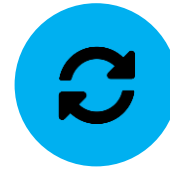
Challenges We Hit — And Solved

Real production-style problems, diagnosed and fixed, not avoided.



Corrupted import

A misconfigured header setting silently scrambled a live table — caught, reverted, fixed at the source.



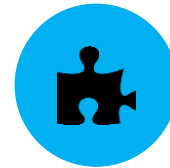
CDC vs. Full Load formats

Incremental update files carry an extra hidden column — pipeline now auto-detects and adapts.



Network access

Dashboard tool couldn't reach the warehouse — resolved by routing traffic securely inside the private network.



A misclassified column

An income category was wrongly read as a number, wiping its data — caught, corrected, verified.

What Makes This Architecture Different

Not just a pipeline — an engineered, cost-aware, production-realistic system.



One engine, four products

A single reusable pipeline handles all product lines — not four separate ones.



Nothing hidden

Every data issue is flagged and traceable, never silently dropped.



CDC-aware by design

Built to recognize and capture incremental change, not just one-time loads.



Cost-conscious

Serverless, pay-per-use choices throughout — no idle infrastructure billing.



Honest trade-offs

Where shortcuts were taken under deadline, they're documented — not hidden.



Tested like production

Deliberately dirty test data proves the pipeline holds up, not just the happy path.

From Data to Decisions

Four real banking questions, answered instantly.



Cross-Sell

Which customers hold only a savings account — the biggest untapped opportunity.



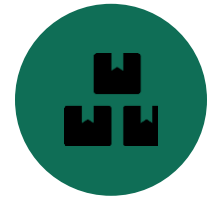
High-Value Clients

Rank customers by total relationship value across all products.



Regional Performance

Compare Products sale across region.



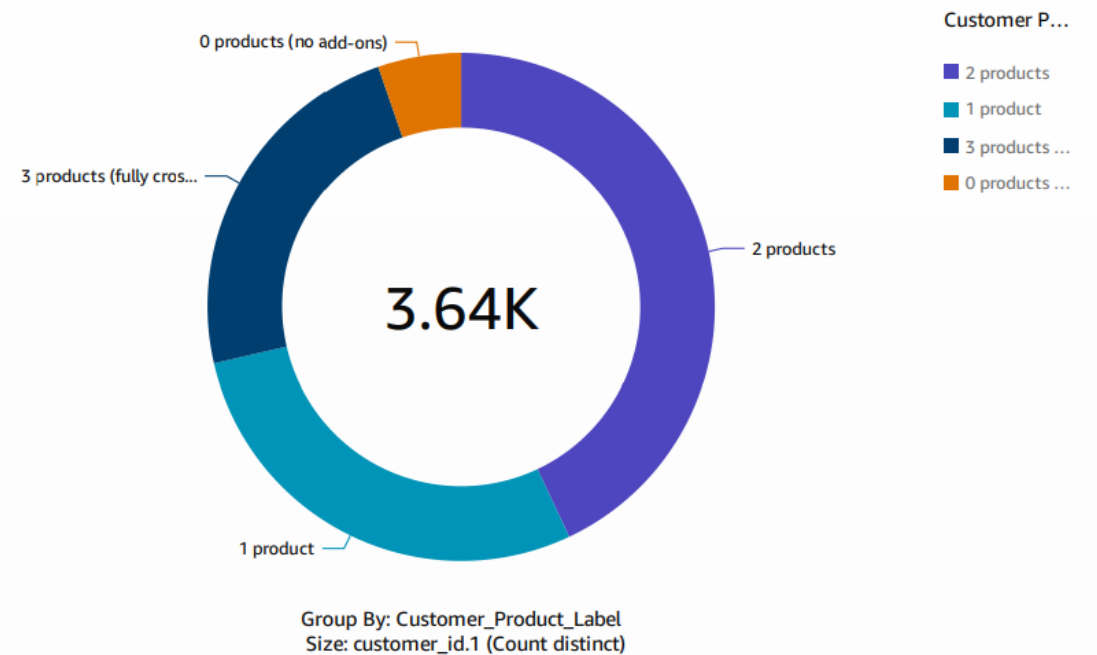
Product Adoption

See which product — Loans, Cards, or Investments — customers choose most.

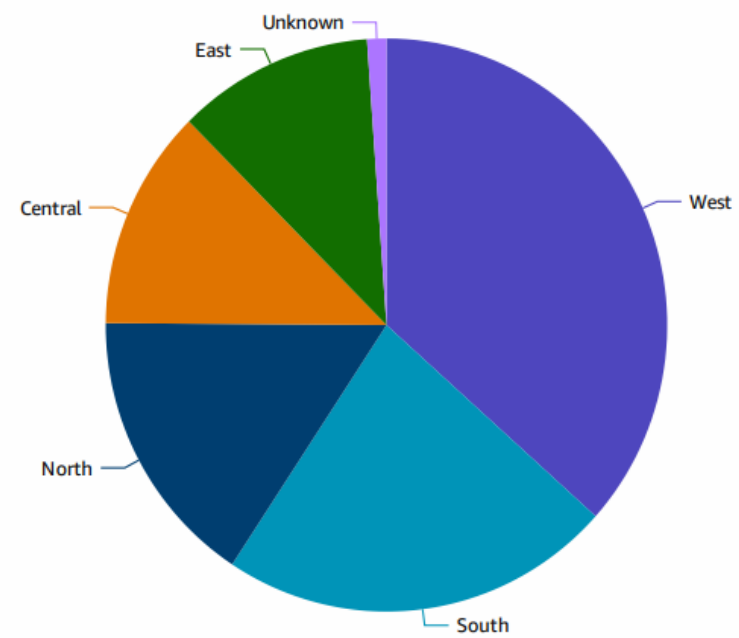
"Top Customers by Relationship Value"

customer_id.1	customer_firstname	customer_lastname	Client_Value
CUST003771	Barbara	Williams	1,866,347.56
CUST002322	Nancy	Miller	1,864,575.43
CUST003760	Elizabeth	Martinez	1,823,653.04
CUST004138	Linda	Rodriguez	1,789,479.53
CUST003053	Betty	Thomas	1,770,728.55
CUST002179	James	Jackson	1,763,615.93
CUST010987	Linda	Taylor	1,711,062.6
CUST004138	Linda	Rodriguez	1,699,553.53
CUST002718	Michael	Johnson	1,697,695.93
CUST003401	James	Wilson	1,694,637.33
CUST003096	Margaret	Anderson	1,685,397.44
CUST010282	Margaret	Rodriguez	1,664,521.59
CUST002607	Elizabeth	Martin	1,661,129.26

"Cross-Sell Opportunity: Products per Customer"

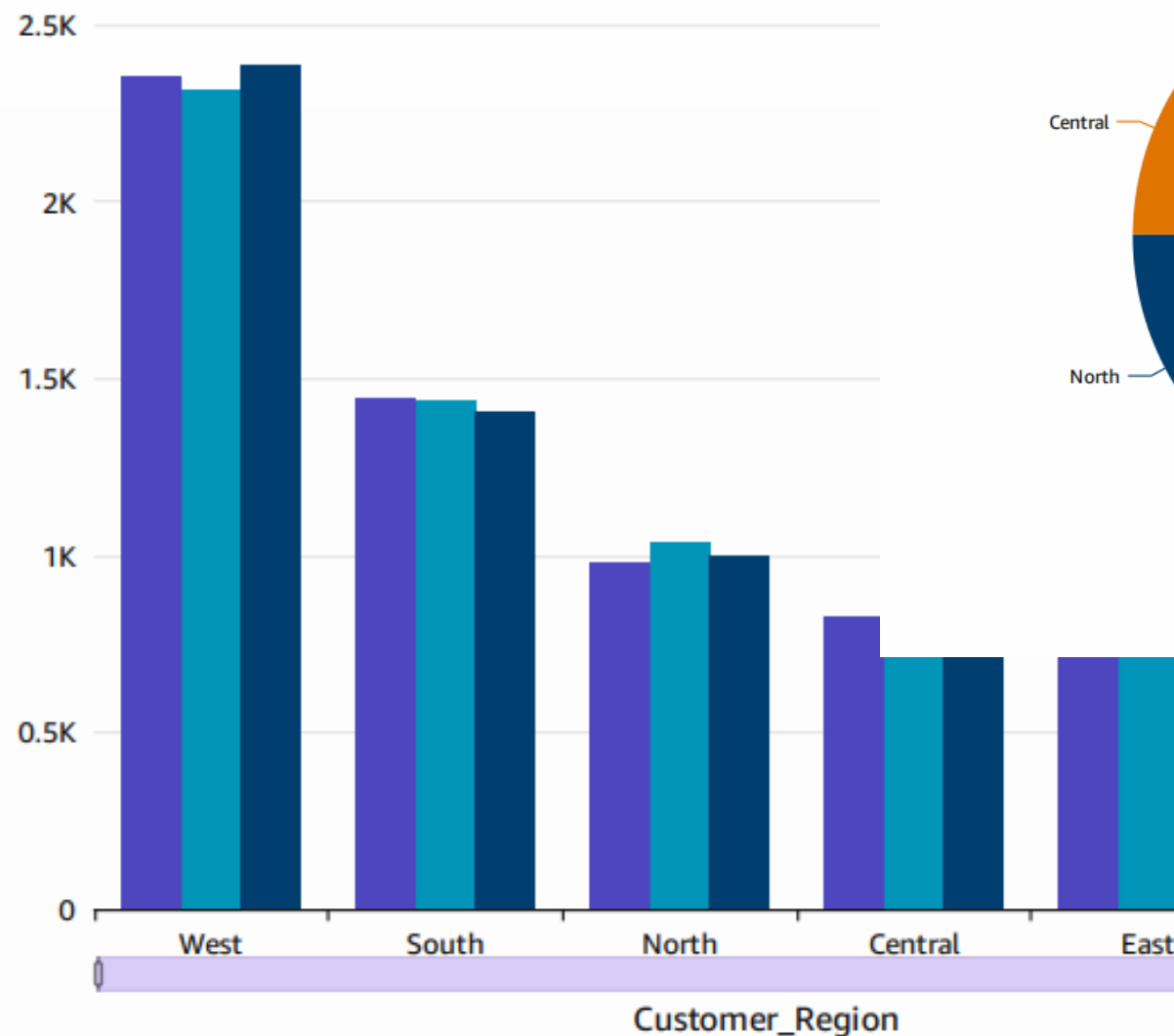


"Customer Distribution by Region"

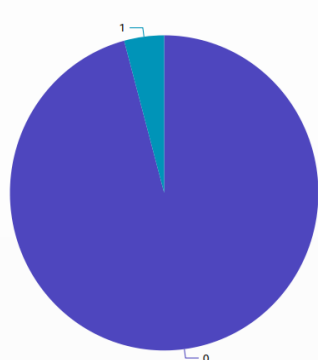


Group By: Customer_Region
Size: customer_id.1 (Count)

"Product Sales by Region"



"Loan Risk Overview: Outstanding vs. Loan Amount"



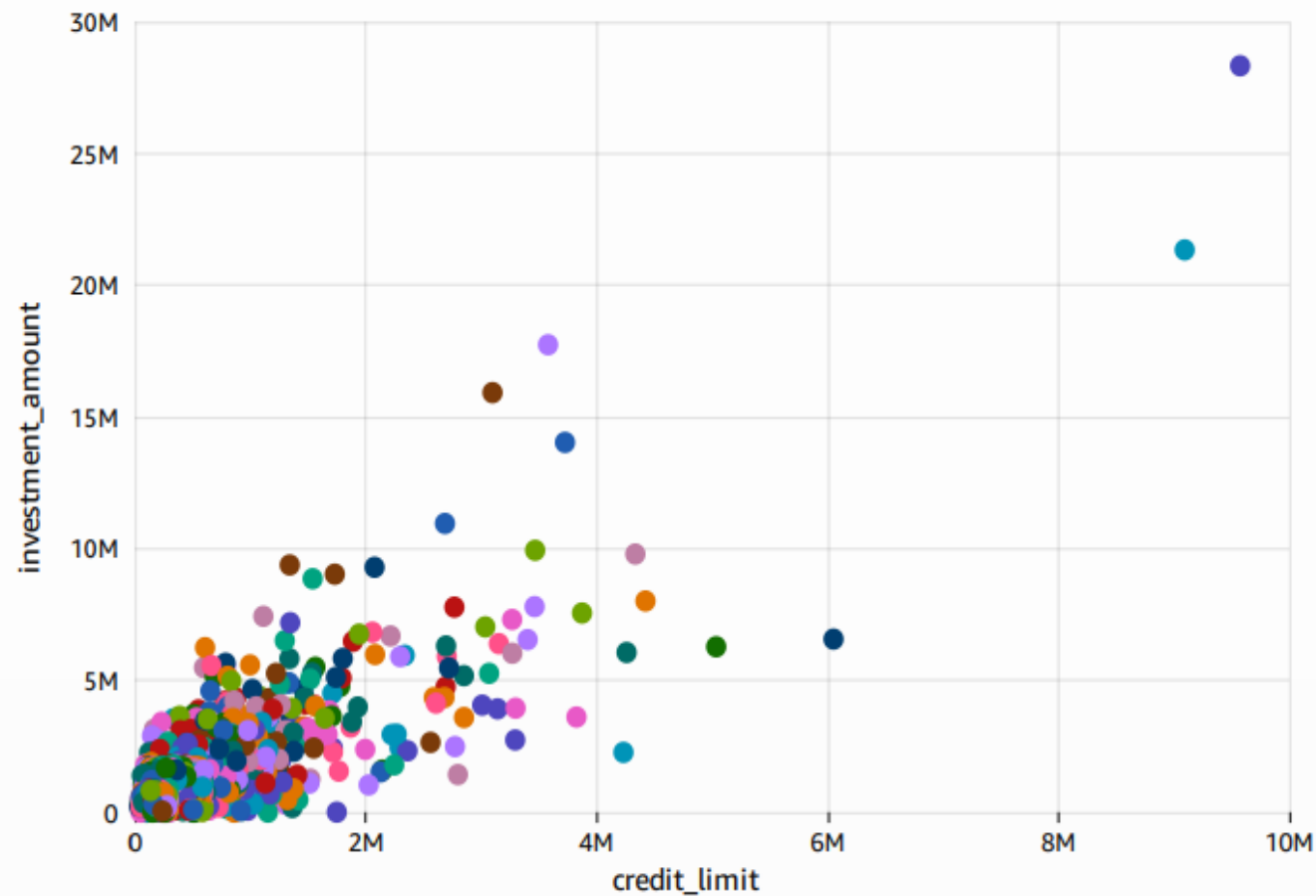
Group By: outstanding_exceeds_loan_flag
Size: loan_id (Count distinct)

Outstandin...

0
1
null

Credit Limit vs. Investment Value

SHOWING UP TO 2500 DATA POINTS



Customer Id 1

CUST002575
CUST010621
CUST002441
CUST003711
CUST002880
CUST003479
CUST000003
CUST000018
CUST010268
CUST003993
CUST010706
CUST010853
CUST000008
CUST002307
CUST002708
CUST002880

What's Next

The roadmap toward a fully self-updating, production-grade platform.



Smart updates (MERGE)

Update only what changed,
instead of reloading
everything.

Full automation

Zero manual steps from data
upload to dashboard refresh.

Native quality rules

Formalize checks using AWS
Glue's built-in data quality
engine.

Self-adapting schema

Automatically detect new
columns as the business
evolves.

Thank You

Questions welcome.

From Ganesh & Priya